

Implementation Plan & Substitution Map

How to replace the US import, function by function — and the phased plan, capex, team and milestones to build it. Gujarat-local direction: graded kraft-sack fibre + cotton + chemistry.

Prepared 2 Oct 2026 · Capex/team figures are indicative planning estimates — confirm with vendors. Proportions validate on the mill trial.

PRINCIPLE The US import is not one thing — it delivers **four jobs at once**: long fibre (RCT), bonding (BF), consistency (grade), and on-tap supply. Substitute **each job** with an India source. Replace the system, not just a material.

1 · Substitution map — what replaces each job of the import

WHAT THE US IMPORT PROVIDES	INDIA SUBSTITUTE	ROLE IN THE BLEND
Long virgin fibre → RCT	Used kraft sacks (cement/chemical/food) + box trim/DLK, graded	Core strength (15–35%)
Extra tear / reinforcement	Cotton waste / linter (Gujarat)	Strength booster (5–8%)
Bulk fibre volume	Domestic OCC + bagasse	Cheap base (50–70%)
Bonding → BF	Cationic starch + GPAM	Wet-end dose
Compression → RCT boost	PVAm + refiner enzyme	Wet-end / refiner
Consistency / graded quality	Your sort → grade → QC to spec process	The value-add & moat
On-tap supply, no 2-mo lock	Local warehouse stock, daily price	The logistics win

Substitution by BF grade (proportions to test)

TARGET	OCC	KRAFT SACKS	COTTON	BAGASSE	CHEMISTRY
18–22 BF	80	10	–	10	starch 1.0% + GPAM 0.2% + enzyme
24–28 BF	60	25	5	10	+ PVAm 0.3%
28–32 BF	50	35	8	7	starch 1.2% + GPAM 0.3% + PVAm 0.4% + enzyme
40–45 BF	low	high	8–10	–	max chemistry; may still need top virgin fibre

Fallback substitutions (flex by price / availability)

IF THIS IS SHORT / COSTLY...	SUBSTITUTE WITH...
Kraft sacks scarce	More box trim / DLK; raise chemistry + enzyme dose
Cotton waste too pricey	Cheaper comber/dropping grades; or more kraft sacks + PVAm
OCC price spikes	More bagasse/straw as filler (short fibre — keep strength from sacks)
Need higher RCT fast	Lift kraft-sack % and PVAm; add toll-pulped cotton/virgin fibre for top grades

Flexibility is the point: because strength comes from a *few* graded long-fibre inputs + chemistry, you can swap the cheap base freely with price — the recipe is a dial, not a fixed formula.

2 · The plan — phase by phase

PHASE 0 · 0–2 MONTHS · VALIDATE

Prove the substitution works

- Buy **one US import bale** + collect local kraft-sack / cotton / OCC samples.
- Lab-test the import → **write the target spec** (fibre length, BF/RCT contribution, cleanliness, moisture).
- Run the **mill trial** (Control vs Mix A/B/C) on one associated mill; measure BF, RCT, colour, smell, runnability, ₹/tonne.
- ✓ Milestone: a passing recipe + proven ₹ saved/tonne. **Cost: low (samples + lab + trial time).**

PHASE 1 · 2-6 MONTHS · SET UP

Build the Fibre Grading & Blending Unit

- Lease a shed/warehouse in the **Morbi/Vapi/Ahmedabad** belt.
 - Install the line: intake, **sorting tables/conveyor**, de-contamination, moisture control, **baler**, weighbridge, small **QC lab** (burst + ring-crush testers).
 - Lock supply contracts for kraft sacks, box trim, cotton waste.
 - **File patent/IP** on the recipe + grading method; register the grade brand ("India-DLK Grade A").
 - Sign first **5-10 mills** on gain-share.
- ✓ Milestone: unit producing graded bales to spec; first paying mills. **Capex: indicative ₹50 L-₹1.5 Cr for a modest unit (confirm with vendors).**

PHASE 2 · 6-18 MONTHS · SCALE GUJARAT

Own the belt

- Roll across the **50 associated mills** + Morbi/Vapi cluster.
 - Build a **recipe library by BF grade**; standardise QC + spec sheets per grade.
 - Scale sorting throughput; add a second shift / baler as volume grows.
 - Formalise gain-share contracts; build the technical-service team.
- ✓ Milestone: steady recurring volume & margin across dozens of mills.

PHASE 3 · 18 MONTHS+ · PAN-INDIA

Expand the brand

- Replicate the unit model in other mill clusters (UP-Muzaffarnagar, Punjab).
 - Scale to **500+ mills**; position "India-DLK" as a known graded brand.
 - Add a **premium tier** (toll-pulped cotton/MFC) for top grades as price allows.
- ✓ Milestone: national graded-fibre brand substituting imports at scale.

3 · Capex, team & KPIs (indicative)

CAPEX ITEM	INDICATIVE	TEAM	ROLE
Shed / warehouse (lease)	monthly	Plant/ops manager	Run the line
Sorting line + conveyor	₹10-40 L	QC chemist	Spec & lot testing
Baler + weighbridge	₹10-30 L	Sorting labour (10-20)	De-contaminate & grade
De-dust / moisture control	₹5-20 L	Technical-service (2-3)	Mill trials & dosing
QC lab (burst, ring-crush, Cobb)	₹5-12 L	Sales + logistics	Contracts & delivery
Total (modest unit)	~₹50 L-₹1.5 Cr		

KPIs to track

METRIC	TARGET
% imported fibre displaced at client mills	30-60%+
Furnish ₹/tonne saved vs import recipe	net positive
Lot pass-rate vs spec (consistency)	> 95%
Grade BF/RCT vs import benchmark	match ±tolerance
Mills on recurring gain-share	grow 5 → 50 → 500

Risks & mitigations: (1) **Smell/drainage** from recovered fibre → hard QC pass/fail + enzyme for drainage. (2) **Supply variability** → multiple sack/trim contracts + fallback substitutions above. (3) **Quality consistency** → the spec + lot QC is exactly the mitigation. (4) **Competitors move down-market** → patent + first-mover channel lock. (5) All figures indicative → validate on trial & with vendors before committing capital.

Companion to: research dossier · full-process & recipe · mill-trial protocol · QC checklist · competitor scan (all in PR #3 / research/). Prices & capex indicative (India 2025) — confirm by quotation.